

Rout Enterprises

Seamless strength,
Endless possibilities...

Building the backbone of modern infrastructure



SEAMLESS PIPES AND TUBES

PRODUCT RANGE

SEAMLESS PIPES & TUBES

Reflects our commitment to delivering pipes and tubes engineered for durability, precision, and reliability in the most demanding environments.

1

**Stainless Steel
Seamless Pipes -
Solution Annealed
& Pickled**

2

**Stainless Steel
Precision & Heat
Exchanger Tubing -
Solution Annealed &
Pickled Or
Bright Annealed**

3

**Hydraulic &
Instrumentation
Tubing -
Bright Annealed**

STAINLESS STEEL SEAMLESS PIPES – SOLUTION ANNEALED & PICKLED

Outside Diameter	1/2" NPS to 6" NPS
Wall Thickness	SCH 10S to SCH 160
Grades	304/304L/1.4301/1.4307,316/316L/1.4401/1.4404, 1.4571/316Ti, 1.4541/321, 1.4878/321H, UNS S32205
Specifications	ASTM, ASME, DIN EN (GERMAN)
*Note: Other Sizes and Thickness can be produced based on Client Requirement.	

STAINLESS STEEL SEAMLESS TUBES (PRECISION & HEAT EXCHANGER TUBING) - SOLUTION ANNEALED & PICKLED OR BRIGHT ANNEALED

Outside Diameter	15.90 MM to 60.3 MM
Grades	304/304L/1.4301/1.4307,316/316L/1.4401/1.4404, 1.4571/316Ti, 1.4541/321, 1.4878/321H, UNS S32205
Specifications	ASTM, ASME, DIN EN (GERMAN), NF (AFNOR), JIS (JAPAN)

****Bright Annealed till 40MM**

PACKING

Pipes are packed in 500 kg to 1000 kg bundles. Each Pipe seal with End Cap then wrapped in HDPE cloth and strapped along with identification labels on each bundle.

TRANSPORTATION MODES:

· Domestic: By road across India · International : By sea or air worldwide

For export orders, we facilitate shipment through: 20', 40', and 45' containers (Open or Closed top)

INSPECTION :

Rout Enterprises pipes and tubes are inspected at advanced in-house manufacturing and testing facilities by qualified personnel, client representatives, and reputed international agencies, in accordance with customer requirements. All materials are certified as per 3.1/3.2 standards based on client specifications.

APPROVALS

Following approvals for seamless pipes and tubes ensure compliance with quality standards, material integrity, and suitability for safe & reliable performance in critical applications.

TUV : ISO9001:2015
TUV : AD 2000-Merkblatt W0
IBR
RINA
Currently Being Processed
BV/ABS/DNV/Loyd's Register

SURFACE

CONDITIONS		
Symbol	Process Route	Surface Condition
CFD	Cold Finished heat treated, Descaled	Metallically Clean
CFA	Cold Finished Bright Annealed	Metallically Bright Annealed
CFC	Cold Finished Solution Annealed, Ground	Metallically Bright Ground, The type of Grinding and Degree of Roughness shall be agreed at the time of enquiry and order
CFP	Cold Finished Solution Annealed, Polished	Metallically Bright Polished, The type of Polishing and Degree of Roughness shall be agreed at the time of enquiry and order

TOLERANCE CHART

DIMENSIONS	OUTSIDE DIAMETERS	WALL THICKNESS
Tolerance According To EN ISO 1127		
D4/T4	±0.50% Or Min. ±0.10 MM Whichever Is Greater	±7.50% Or ±0.15 MM Whichever Is Greater
D3/T3	±0.70% Or Min. ±0.30 MM Whichever Is Greater	±10% Or ±0.20 MM Whichever Is Greater
Tolerance According To ASTM A312/ASTM A 999		
> 10.30 < 48.30	OD. +0.40 MM/-0.80 MM	THK. -12.50%
> 48.30 < 113.30	OD. +0.80 MM/-0.80 MM	THK. -12.50%
> 114.30 < 219.10	OD. +1.60 MM/-0.80 MM	THK. -12.50%
Tolerance According ASTM 213/ASTM A 1016		
> 10.30 < 48.30	OD. +0.10 MM/-0.11 MM	THK. -0.00%/+20.0%
> 25.40 < 38.10	OD. +0.15 MM/-0.15 MM	THK. -0.00%/+20.0%
> 38.10 < 50.80	OD. +0.20 MM/-0.20 MM	THK. -0.00%/+20.0%
> 50.80 < 63.50	OD. +0.25 MM/-0.25 MM	THK. -0.00%/+20.0%
> 63.50 < 76.20	OD. +0.30 MM/-0.30 MM	THK. -0.00%/+20.0%
> 76.20 < 101.60	OD. +0.38 MM/-0.38 MM	THK. -0.00%/+20.0%
> 101.60	OD. +0.38 MM/-0.64 MM	THK. -0.00%/+20.0%

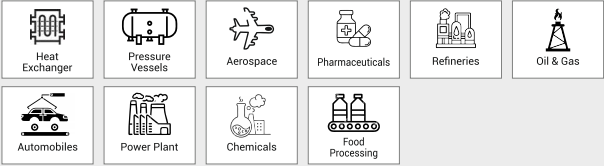
STAINLESS STEEL SEAMLESS TUBES (HYDRAULIC & INSTRUMENTATION TUBING) - BRIGHT ANNEALED

Outside Diameter	12.00 MM to 40.00 MM
Grades	304/304L/1.4301/1.4307, 316/316L/1.4401/1.4404, 1.4571/316Ti, 1.4878/321H, UNS S32205
Specifications	ASTM, ASME, DIN EN (GERMAN), NF (AFNOR), JIS (JAPAN)

(Can offer May 2026 Onward)



APPLICATIONS :



GRADES

Material	ASTM Grades	UNS Grades	DIN EN Grades	JIS Grades
Austenitic	TP 304	S30400	1.4301	SUS304TB
	TP 304L	S30403	1.4306	-
	TP 304L	S30403	1.4307	SUS304LTB
	TP 316	S31600	1.4401	SUS316TB
	TP 316L	S31603	1.4404	SUS316LTB
	TP 316H	S31609	1.4918	SUS316HTB
	TP 316Ti	S31635	1.4571	SUS316TiTB
	TP 321	S32109	1.4541	SUS321TB
Duplex	TP 321H	S32109	1.4941	SUS321HTB
	2205	UNS S32205	1.4462	-

Seamless pipes and tubes are manufactured without welds, offering superior strength, uniformity, and resistance to pressure. Each grade is selected based on mechanical properties, corrosion resistance, temperature tolerance, and application needs.

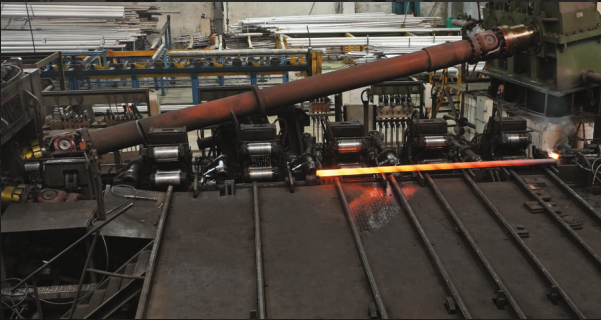
SPECIFICATIONS

EUROPEAN STANDARD	
DIN EN 10216-5	Specification for Seamless Steel Tubes for Pressure Purpose
DIN EN 10217-7	Specification for Welded Steel Tubes for Pressure Purpose
DIN EN 10297-2	Specification for Seamless Steel Tubes for Mechanical and General engineering Purpose
DIN EN 10305-1	Specification for Steel Tubes for Precision Application

ASTM/ASME STANDARD	
A213/SA213	Specification for Seamless Ferretic and Austenitic Alloy Steel Boiler, Super heater and Heat Exchanger Tubes
A249/SA 249	Specification for Welded Austenitic Steel Boiler, Super heater, heat Exchanger and Condenser Tubes
A 268/SA 268	Seamless and Welded Ferritic and Martensitic Stainless Steel Tubing for General Service
A 269	Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service
A 312/SA 312	Specification for Seamless and Welded Austenitic Steel Pipes
A 376/SA 376	Specification for Seamless Austenitic Steel Pipe for High Temperature Service
A 688/ SA 688	Specification for Seamless and Welded Austenitic Stainless Steel Feed water Heater Tubes
A 789/SA 789	Specification for Seamless and Welded Ferritic /Austenitic Stainless Steel Tubing for General Services
A 790/SA 790	Specification for Seamless and Welded Ferritic/Austenitic Stainless Steel Pipes
A 999/SA 999	Specification for General requirement for Alloys and Stainless Steel Pipes
A 1016/SA 1016	Specification for General requirement for Ferritic Alloys Steel, Austenitic Alloys Steel and Stainless Steel Tubes

GERMAN STANDARD	
DIN 11850	Specification for Stainless Steel Tubes for the Food and Chemical industries - Dimensions, Material.
DIN 17455	Specification for General Purpose Welded Circular Stainless Steel Tubes
DIN 17456	Specification for General Purpose Seamless Circular Stainless Steel Tubes
DIN 17457	Specification for Welded Circular austenitic Stainless Steel tubes Subject to Special requirement
DIN 17458	Specification for Seamless Circular austenitic Stainless Steel tubes Subject to Special requirement
DIN 28180	Specification for Seamless Steel Tubes for Heat Exchanger
DIN 11850	Specification for Welded Tubes and Pipes for Food, Beverages, Chemical & Pharmaceutical industry.

RUSSIAN STANDARD	
GOST 9941	Specification for Seamless and Warm -Deformed Tubes made from Corrosion - Resistant Steel



QUALITY CONTROL

quality is our top priority. We ensure flawless products through stringent quality measures and world-class lab testing. Committed to excellence, we leave no stone unturned in meeting customer expectations. Our rigorous approach includes two key testing processes to guarantee top-tier delivery and service.

Non-Destructive Testing

Hydro Test	Air Under Water Test
Eddy Current Test	Dye Penetrant Test
Ultrasonic Test	Baroscopic Inspection
P.M.I Test	Radioactive Contamination Checking
Spectro Testing	Surface Roughness Test

Destructive Testing

Tensile Test	Flange Test
Impact Test	IGC Test – Practice A,B,C & E
Hardness Testing	Impact Test
Flattening Test	Reverse Flattening Test
Flaring Test	

